

Réunion d'avancement n° 15

Algorithmes de navigation par champ magnétique terrestre
pour des véhicules aérospatiaux

22/10/2025

RKHS : le retour

- Espace de Hilbert $\mathcal{F} \subset \Omega^{\mathbb{R}}$ doté d'un noyau $k \in \mathcal{F}$
- Produit scalaire $\langle \cdot | \cdot \rangle_{\mathcal{F}}$ vérifiant $f(x) = \langle f | k(x, \cdot) \rangle_{\mathcal{F}}$

- Feature map (fonction de caractéristiques)

$$\phi(x) := k(x, \cdot) : x' \mapsto k(x, x')$$

- Mean embedding (plongement dans l'espace des caractéristiques)

$$\begin{aligned}\mu_X &:= \mathbb{E}_X(\phi(X)) = \int_{\Omega} \phi(x) dP(x) \\ \Rightarrow \langle f | \mu_X \rangle_{\mathcal{F}} &= \mathbb{E}(f(X))\end{aligned}$$

RKHS : plongement dans l'espace des caractéristiques

- Distributions jointes :

$$C_{X Y} := \mathbb{E}_{X Y}(\phi(X) \otimes \phi(Y)) = \int_{\Omega \times \Omega} \phi(x) \otimes \phi(y) dP(x, y)$$

- Distribution conditionnelle :

$$\mu_{Y|x} := \mathbb{E}_{Y|x}(\phi(Y)) = \int_{\Omega} \phi(y) dP(y|x)$$

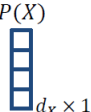
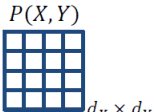
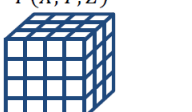
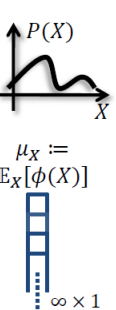
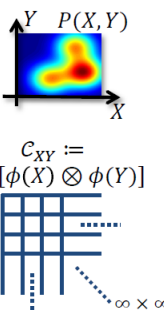
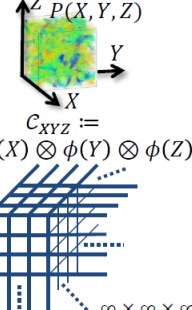
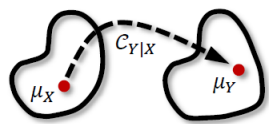
- on veut définir $C_{Y|X}$ vérifiant $\mu_{Y|x} = C_{Y|X} \phi(x)$, avec

$$C_{Y|X} = C_{Y X} (C_{X X} + \lambda I)^{-1}$$

- pour obtenir $\langle f | \mu_{Y|x} \rangle_{\mathcal{F}} = \mathbb{E}_{Y|x}(f(Y))$?

RKHS : estimation sur des données d'entraînement

- pour des données $\mathcal{D}_{XY} := \{(x_1, y_1), \dots, (x_n, y_n)\}$
- $\mu_X \simeq \hat{\mu}_X := \frac{1}{n} \sum_{i=1}^n \phi(x_i) = \Phi(K + \lambda I)^{-1} Y^T$
- $C_{XY} \simeq \hat{C}_{XY} := \frac{1}{n} \sum_{i=1}^n \phi(x_i) \otimes \phi(y_i) = \Phi(K + \lambda I)^{-1} K_{:x}$
- d'autres propriété pour changer de loi *a priori* :

	Distributions			Probabilistic Operations
Discrete	 $P(X)$ $d_X \times 1$	 $P(X, Y)$ $d_X \times d_Y$	 $P(X, Y, Z)$ $d_X \times d_Y \times d_Z$	Sum Rule: $Q(X) = \sum_Y P(X Y)\pi(Y)$ Product Rule: $Q(X, Y) = P(X Y)\pi(Y)$ Bayes Rule: $Q(Y x) = \frac{P(x Y)\pi(Y)}{Q(X)}$
Kernel Embedding	 $P(X)$ $\mu_X := \mathbb{E}_X[\phi(X)]$ $\infty \times 1$	 $P(X, Y)$ $C_{XY} := \mathbb{E}_{XY}[\phi(X) \otimes \phi(Y)]$ $\infty \times \infty$	 $P(X, Y, Z)$ $C_{XYZ} := \mathbb{E}_{XYZ}[\phi(X) \otimes \phi(Y) \otimes \phi(Z)]$ $\infty \times \infty \times \infty$	 Sum Rule: $\mu_X^\pi = C_{X Y} \mu_Y^\pi$ Product Rule: $C_{XY}^\pi = C_{X Y} C_{Y X}^\pi$ Bayes Rule: $\mu_{Y x}^\pi = C_{Y x}^\pi \phi(x)$

Filtre de Kalman dans un RKHS

Algorithm 1: The Kernel Kalman Filter

input: triples $\{(\tilde{x}_1, x_1, y_1), \dots, (\tilde{x}_m, x_m, y_m)\}$,
kernel functions k and g , regularization parameters λ and κ ,
let $\mathbf{X}$ be the matrix of all x_i as columns, $\tilde{\mathbf{X}}$ and $\mathbf{Y}$ analogously,
let $\mathbf{X}_0$ be the matrix of all data points used to compute the initial embeddings

compute kernel matrices

$\mathbf{K} = k(\mathbf{X}, \mathbf{X})$, $\mathbf{K}_{\tilde{\mathbf{X}}\tilde{\mathbf{X}}} = k(\tilde{\mathbf{X}}, \tilde{\mathbf{X}})$, $\mathbf{K}_{\tilde{\mathbf{X}}\mathbf{X}} = k(\tilde{\mathbf{X}}, \mathbf{X})$, and $\mathbf{G} = g(\mathbf{Y}, \mathbf{Y})$

compute model matrices

$\mathbf{T} = (\mathbf{K}_{\tilde{\mathbf{X}}\tilde{\mathbf{X}}} + \lambda \mathbf{I})^{-1} \mathbf{K}_{\tilde{\mathbf{X}}\mathbf{X}}$ and $\mathbf{O} = (\mathbf{K} + \lambda \mathbf{I})^{-1} \mathbf{K}$

compute initial embeddings

kernel matrix with samples of the initial distribution: $\mathbf{K}_0 = k(\mathbf{X}, \mathbf{X}_0)$, $\mathbf{C}_0 = (\mathbf{K} + \lambda \mathbf{I})^{-1} \mathbf{K}_0$,
compute mean and variance over the columns: $\mathbf{m}_0 = \text{mean}(\mathbf{C}_0)$, $\mathbf{S}_0 = \text{var}(\mathbf{C}_0)$

loop

if new observation $\mathbf{y}_t$ available **then**

compute kernel Kalman gain

$\mathbf{Q}_t = \mathbf{S}_t^- \mathbf{O}^\top (\mathbf{G} \mathbf{O} \mathbf{S}_t^- \mathbf{O}^\top + \kappa \mathbf{I})^{-1}$

innovation update

$\mathbf{m}_t^+ = \mathbf{m}_t^- + \mathbf{Q}_t (\mathbf{g}_{:\mathbf{y}_t} - \mathbf{G} \mathbf{O} \mathbf{m}_t^-)$

$\mathbf{S}_t^+ = \mathbf{S}_t^- - \mathbf{Q}_t \mathbf{G} \mathbf{O} \mathbf{S}_t^-$

transition update

$\mathbf{m}_{t+1}^- = \mathbf{T} \mathbf{m}_t^+$, $\mathbf{S}_{t+1}^- = \mathbf{T} \mathbf{S}_t^+ \mathbf{T}^\top + \mathbf{V}$

project into state space

$\boldsymbol{\eta}_t = \mathbf{X} \mathbf{O} \mathbf{m}_t$, $\boldsymbol{\Sigma}_t = \mathbf{X} \mathbf{O} \mathbf{S}_t \mathbf{O}^\top \mathbf{X}^\top$
